

Energy Efficiency

When energy is changed some of it is always wasted.

We can measure the amount of useful energy we get by the efficiency.

$$\text{Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$$

The examples below all show energy changes in engines. The **work** that they do is their 'useful energy out'.

Calculate their efficiencies.

1. A petrol engine takes 1000J of chemical energy and does 250J of work.

2. A diesel engine uses 6000J of chemical energy and does 2100J of work.

3. An electric motor uses 50J of electrical energy and does 40J of work.

4. A cyclist (a human engine) has 200J of energy and does 30J of work.

The following questions are harder, you will need to decide what form the **useful energy** is in.

5. A rocket engine takes in 800J of chemical energy and changes this into 480J of kinetic energy and 320J of heat energy.

6. A jet engine gas turbine takes in 1200J of chemical energy and gives out 960J of kinetic, 180J of heat and 60J of sound energy.

7. A TV takes in 600J of electrical energy and gives out 300J of light, 240J of sound and 60J of heat energy (be careful here!).



